BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 102 · Hexadecimal

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why does 11111111 plus 00000001 overflow eight bits?

Q2. A has code 65. Capital-letter codes go in order. What is D's code?
A. 65
B. 68
C. 67
D. 69

Q3. “Unicode supports fewer characters than standard ASCII.” Correct this. Explain why.

Q4. A program calculates an area and shows a menu in several places. Correct both claims. Explain why. (a) A function cannot return a value. (b) A procedure must return a value.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Hexadecimal — Base-16 representation using 0–9 and A–F.

Nibble — A group of four bits.

Most significant digit — The digit with the highest positional weight.

Denary — Base-10 representation.

Place value — The value assigned to a digit's position.

New learning

Hex uses base 16: Hexadecimal digits are 0–9 and A–F, where A=10 through F=15.

Four bits per digit: One hexadecimal digit represents exactly four binary bits (one nibble).

Binary to hexadecimal: Split an eight-bit value into two four-bit groups and convert each group to one hex digit.

Hexadecimal to denary: Multiply each hex digit by its positional value, 16 and 1, then add the results.

Worked example

Convert 10101111 from binary into hexadecimal and then denary. [5 marks]

1. Group four bits

2. 1010 = 10 = A and 1111 = 15 = F.

3. Write AF in hexadecimal.

4. In denary, A×16+F = 10×16+15 = 175. [5]

We Do · Practise with guidance

Guided 1

Convert binary 11001010 to hexadecimal. [3]

Worked method: 1100 = C [1], 1010 = A [1], so CA [1].

Guided 2

Convert 3E hexadecimal into binary. [3]

Worked method: 3 = 0011 [1] and E = 1110 [1], so 00111110 [1].

Guided 3

Convert 2F hexadecimal to denary. [4]

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The true sum is 256, requiring nine bits; only eight stored bits would contain 00000000.

Q2. B — 68

Q3. Unicode supports far more characters than standard ASCII.

Q4. (a) A function returns a value, such as an area. (b) A procedure performs a task, such as showing a menu, without returning a value. Use arguments and return values to avoid unwanted changes to global data.

I Do / We Do pages · Guided practice

Q1. 1100 = C [1], 1010 = A [1], so CA [1].

Q2. 3 = 0011 [1] and E = 1110 [1], so 00111110 [1].

Q3. 2×16 [1] + 15×1 [1] = 32+15 [1] = 47 [1].

You Do pages · Code First

```
first = 2
second = 11
value = first * 16 + second
print(value)
```

Expected result: 43

You Do pages · Independent questions

Q1. 1001=9 [1], 0110=6 [1], combined hex 96 [1].

Q2. B=1011 [1], 4=0100 [1], so 10110100 [1].

Q3. 7×16 [1] +13 [1] =112+13 [1] =125 [1].

Exit Ticket · Model answer

True

Try the question again

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Worked method: 2×16 [1] + 15×1 [1] = 32+15 [1] = 47 [1].

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the denary calculation for hexadecimal 2B.

```
first = 2
second = 11
value = ...
print(value)
```

write

Question 1

Convert binary 10010110 to hexadecimal. [3] Source: Original OCR J277-style practice question.

Question 2

Convert hexadecimal B4 into eight-bit binary. [3] Source: Original OCR J277-style practice question.

Question 3

Convert hexadecimal 7D to denary. [4] Source: Original OCR J277-style practice question.

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

A trace table can help check how values change during a program.